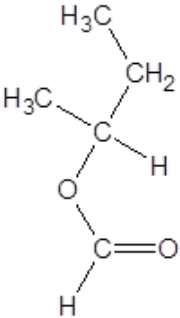


| Question |     |       |  | Marking details                                                                                                                                                                                                                 | Marks available |     |     |       |       |      |
|----------|-----|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9        | (a) | (i)   |  | reflux (1)<br><br>to avoid loss of reagents / products / solvent (1)                                                                                                                                                            | 2               |     |     | 2     |       | 2    |
|          |     | (ii)  |  | distillation (1)<br><br>boiling temperature of ester is lower than alcohol/acid (1)                                                                                                                                             | 2               |     |     | 2     |       | 2    |
|          |     | (iii) |  | (conc) sulfuric acid                                                                                                                                                                                                            | 1               |     |     | 1     |       | 1    |
|          |     | (iv)  |  | reagents<br><br>$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3 + \text{HCOOH}$ (1)<br><br>products<br><br> + $\text{H}_2\text{O}$ (1) |                 | 2   |     | 2     |       |      |

| Question                           |                                          |            |                        | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks available |                                          |            |                        |                        |                                 |                                   |   |   |   |    |                                   |   |   |   |                                |                 |   |   |   |                                    |                      |   |   |   |                             |                    |   |   |   |  |   |  |   |  |   |
|------------------------------------|------------------------------------------|------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------|------------|------------------------|------------------------|---------------------------------|-----------------------------------|---|---|---|----|-----------------------------------|---|---|---|--------------------------------|-----------------|---|---|---|------------------------------------|----------------------|---|---|---|-----------------------------|--------------------|---|---|---|--|---|--|---|--|---|
|                                    |                                          |            |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2                                      | AO3        | Total                  | Maths                  | Prac                            |                                   |   |   |   |    |                                   |   |   |   |                                |                 |   |   |   |                                    |                      |   |   |   |                             |                    |   |   |   |  |   |  |   |  |   |
|                                    | (b)                                      |            |                        | <p>award (1) each for any <b>two</b> of following</p> <ul style="list-style-type: none"><li>• Na<sub>2</sub>CO<sub>3</sub> / Mg - effervescence / fizzing / bubbles</li><li>• acidified potassium dichromate - orange to green<br/>accept acidified potassium manganate(VII) - purple to colourless</li><li>• iodine and sodium hydroxide / potassium iodide and sodium chlorate(I)<br/>- yellow precipitate</li></ul> <p>award (1) for correct ✓/× for all three compounds for first test<br/>award (1) for correct ✓/× for all three compounds for second test</p> <p>award (1) for tests that enable all three compounds to be identified</p> <table><tr><th>Reagent(s)</th><th>Observation expected for positive result</th><th>butan-2-ol</th><th>2-methylpropanoic acid</th><th>3-hydroxybutanoic acid</th></tr><tr><td>Na<sub>2</sub>CO<sub>3</sub></td><td>effervescence / fizzing / bubbles</td><td>×</td><td>✓</td><td>✓</td></tr><tr><td>Mg</td><td>effervescence / fizzing / bubbles</td><td>×</td><td>✓</td><td>✓</td></tr><tr><td>acidified potassium dichromate</td><td>orange to green</td><td>✓</td><td>×</td><td>✓</td></tr><tr><td>acidified potassium manganate(VII)</td><td>purple to colourless</td><td>✓</td><td>×</td><td>✓</td></tr><tr><td>iodine and sodium hydroxide</td><td>yellow precipitate</td><td>✓</td><td>×</td><td>✓</td></tr></table> | Reagent(s)      | Observation expected for positive result | butan-2-ol | 2-methylpropanoic acid | 3-hydroxybutanoic acid | Na <sub>2</sub> CO <sub>3</sub> | effervescence / fizzing / bubbles | × | ✓ | ✓ | Mg | effervescence / fizzing / bubbles | × | ✓ | ✓ | acidified potassium dichromate | orange to green | ✓ | × | ✓ | acidified potassium manganate(VII) | purple to colourless | ✓ | × | ✓ | iodine and sodium hydroxide | yellow precipitate | ✓ | × | ✓ |  | 5 |  | 5 |  | 5 |
| Reagent(s)                         | Observation expected for positive result | butan-2-ol | 2-methylpropanoic acid | 3-hydroxybutanoic acid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |                                          |            |                        |                        |                                 |                                   |   |   |   |    |                                   |   |   |   |                                |                 |   |   |   |                                    |                      |   |   |   |                             |                    |   |   |   |  |   |  |   |  |   |
| Na <sub>2</sub> CO <sub>3</sub>    | effervescence / fizzing / bubbles        | ×          | ✓                      | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 |                                          |            |                        |                        |                                 |                                   |   |   |   |    |                                   |   |   |   |                                |                 |   |   |   |                                    |                      |   |   |   |                             |                    |   |   |   |  |   |  |   |  |   |
| Mg                                 | effervescence / fizzing / bubbles        | ×          | ✓                      | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 |                                          |            |                        |                        |                                 |                                   |   |   |   |    |                                   |   |   |   |                                |                 |   |   |   |                                    |                      |   |   |   |                             |                    |   |   |   |  |   |  |   |  |   |
| acidified potassium dichromate     | orange to green                          | ✓          | ×                      | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 |                                          |            |                        |                        |                                 |                                   |   |   |   |    |                                   |   |   |   |                                |                 |   |   |   |                                    |                      |   |   |   |                             |                    |   |   |   |  |   |  |   |  |   |
| acidified potassium manganate(VII) | purple to colourless                     | ✓          | ×                      | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 |                                          |            |                        |                        |                                 |                                   |   |   |   |    |                                   |   |   |   |                                |                 |   |   |   |                                    |                      |   |   |   |                             |                    |   |   |   |  |   |  |   |  |   |
| iodine and sodium hydroxide        | yellow precipitate                       | ✓          | ×                      | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 |                                          |            |                        |                        |                                 |                                   |   |   |   |    |                                   |   |   |   |                                |                 |   |   |   |                                    |                      |   |   |   |                             |                    |   |   |   |  |   |  |   |  |   |
|                                    |                                          |            |                        | Question 9 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5               | 7                                        | 0          | 12                     | 0                      | 10                              |                                   |   |   |   |    |                                   |   |   |   |                                |                 |   |   |   |                                    |                      |   |   |   |                             |                    |   |   |   |  |   |  |   |  |   |